

PROCESS THERMAL MANAGEMENT SYSTEMS • WCP SERIES

Air & Water-Cooled Industrial Process Chillers

High-Precision Closed-Loop Water Chillers with Integrated SS304 Buffer Reservoir & Centrifugal Pumps

Precision $\pm 0.5^{\circ}\text{C}$ Temperature Regulation ($+5^{\circ}\text{C}$ to $+25^{\circ}\text{C}$): Designed for continuous 24/7 duty without compressor hunting. The oversized SS304 internal reservoir dampens sudden thermal spikes from molding cycles, laser piercing, or spindle speed changes.

Win Equipments WCP Series chillers integrate Copeland and Danfoss hermetic scroll compressors, non-ferrous SS304 wetted plumbing, high-efficiency plate heat exchangers, and digital PID controllers. Available in stand-alone air-cooled and cooling-tower coupled water-cooled configurations.



SCROLL COMPRESSOR

$\pm 0.5^{\circ}\text{C}$ PRECISION

SS304 BUFFER TANK

ECO R410A / R407C

0.5–150 TR

COOLING RANGE

$\pm 0.5^{\circ}\text{C}$

CONTROL STABILITY

15–700 L

SS304 TANK VOLUME

4.5+ COP

THERMODYNAMIC COP

CORE ENGINEERING ADVANTAGES

Copeland / Danfoss Scroll

High-efficiency scroll compression elements offer 70% fewer moving parts, high volumetric efficiency, and quiet operation below 68 dBA.

SS304 Non-Ferrous Loop

Evaporator coils, buffer reservoir, and centrifugal pump impellers are non-ferrous stainless steel, preventing galvanic rust and debris clogging.

Hot Gas Bypass Circuit

Modulating hot gas bypass valve balances system capacity down to 10% load, maintaining setpoint stability without damaging compressor short-cycling.

TARGET INDUSTRIAL APPLICATIONS

Plastic Injection Molding & Extrusion

Fiber Laser Resonators & Cutting Optics

High-Speed CNC Electro-Spindles

Chemical Reaction Vessel Jacketing

Beverage, Brewery & Dairy Processing

Pharmaceutical Formulations

ENGINEERING SPECIFICATIONS MATRIX

WCP Series Technical Data (0.5 to 50 TR)

Standard Ratings: 12°C leaving water temperature, 7°C delta T, 35°C ambient air temperature

MODEL NO.	CAPACITY	COOLING	COMP. POWER	ELECTRICAL SUPPLY	WATER FLOW	PUMP	TANK	CONNECTION
Model Code	TR	kW	kW	V / Ph / Hz	LPM	HP	Liters	BSP / Flange
WCP 005	0.5 TR	1.7 kW	0.9 kW	230V / 1Ph / 50Hz	6 LPM	0.5 HP	15 L	3/4" BSP
WCP 010	1.0 TR	3.5 kW	1.5 kW	230V / 1Ph / 50Hz	10 LPM	0.5 HP	25 L	3/4" BSP
WCP 020	2.0 TR	7.0 kW	2.5 kW	230V / 1Ph / 50Hz	21 LPM	0.75 HP	40 L	1" BSP
WCP 030	3.0 TR	10.5 kW	3.4 kW	415V / 3Ph / 50Hz	32 LPM	1.0 HP	60 L	1" BSP
WCP 050	5.0 TR	17.5 kW	6.0 kW	415V / 3Ph / 50Hz	52 LPM	1.5 HP	80 L	1.25" BSP
WCP 075	7.5 TR	26.3 kW	8.5 kW	415V / 3Ph / 50Hz	78 LPM	2.0 HP	120 L	1.5" BSP
WCP 100	10.0 TR	35.1 kW	11.2 kW	415V / 3Ph / 50Hz	105 LPM	3.0 HP	160 L	2" BSP
WCP 150	15.0 TR	52.7 kW	16.5 kW	415V / 3Ph / 50Hz	158 LPM	3.0 HP	250 L	2" BSP
WCP 200	20.0 TR	70.3 kW	22.0 kW	415V / 3Ph / 50Hz	210 LPM	5.0 HP	350 L	2.5" BSP
WCP 300	30.0 TR	105.5 kW	33.0 kW	415V / 3Ph / 50Hz	315 LPM	7.5 HP	450 L	3" Flange
WCP 500	50.0 TR	175.8 kW	54.0 kW	415V / 3Ph / 50Hz	525 LPM	10.0 HP	700 L	4" Flange

Hydraulic Integration Note: Dual independent refrigeration circuits are standard on models 15 TR and above for N+1 operational redundancy. Water-cooled shell & tube condenser models connect directly to Win FRP cooling towers.

Request Technical Submittal & Commercial Quotation

Direct factory engineering consultation for custom capacity sizing, P&ID schematics & dimensional drawings.

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